

construction / reconstruction of the self is tantamount to managing the changing set of flows and codes that people are confronted with in their daily existence' (Castells 1996b: 34). Or, as he puts it in an autobiographical aside, 'I am afraid I look more like a flow' (Castells 1996a / 2000: 243).

SUMMARY

The network society has a number of distinguishing features, though critics are unsure just how 'new' this all is (Castells says the question of newness is boring). We are now living under the condition of global informational capitalism, brought about by three independent processes: the information technology 'revolution', the crisis and restructuring of capitalism and statism, and the rise of new social movements. Out of this whirl of change come the network society and the network logic. The global economy is chaotic and turbulent, and seems to function autonomously, with very variable impacts around the world. There is a new spatial division of labour, based around self-programmable elite labour, replaceable generic labour, and the discarded jobless. Working now means networking, and organizations become network enterprises while individuals develop portfolio careers or risk being switched off. The elites of self-programmable labour live exclusive lifestyles while social exclusion and poverty escalate around them; the poor and excluded react by joining the global criminal networks, or by withdrawing into fundamentalisms. Networking logic extends to the self, either by counterposing the net and the self or through network individualism, though grassroots offers ways to bring together the net and the self, through the networking of project identities.

2 SPACE OF FLOWS

Castells says that his concept of the space of flows is very important and very complex, even counter-intuitive. First elaborated in *The Informational City*, it is one area of *The Rise of the Network Society* where he permits himself a more theoretical approach (and later even admits to previous theoretical mistakes; see Castells and Ince 2003). Based on the observation that 'dominant functions were increasingly operating on the basis of exchanges between electronic circuits linking up information systems in different locations', Castells' central premise is the dichotomy between the space of flows

and the space of places as the geographical manifestations of the network society (Castells 2002 [1996]: 131). The space of flows is woven closely together with his idea of ‘timeless time’, which I also discuss in this section.

NETWORKS AND FLOWS

The notion of flows is integral to the idea of the network: ‘the space of flows is the material organization of time-sharing social practices that work through flows’ (Castells 2002 [1996]: 132). The network is a series of points, hubs or nodes – these can be people, cities, businesses, nationstates – connected together by flows of various sorts: flows of information, of materials, of money, of people. So the space of flows is both the nodes and the connecting flows. Hence part of its complexity. Now, the idea of global flows had already been discussed lucidly by Arjun Appadurai (1990), who is not referred to by Castells but whose ideas nevertheless have a striking resemblance. Trying to think about globalization, Appadurai conceptualized the world being criss-crossed by a series of disjunctive or non-isomorphic flows (see p. 62). He listed five, and gave each the suffix -scape, to conjure images of a landscape, all hills and valleys, humps and bumps: ethnoscap, finanscap, technoscap, ideoscap and mediascap (respectively, flows of people, money, technology and skill, ideologies, and media texts). These flows are chaotic, varying in speed and intensity, overlapping, attracting or repelling one another. Moreover, they don’t merely circle the globe like satellites; they land. Where they land, and what results from their landing, is similarly unpredictable (even though there are patterns and there is stability in parts). Where they land are the nodes in the network. But there are different kinds of nodes, some with greater capacity to control at least some flows, for example by refusing entry to some people, or blocking some media content from being accessible. ‘Strong’ nodes can act as magnets, drawing down some flows while also pushing others away, deflecting them elsewhere. Other nodes are relatively powerless, and are endlessly buffeted and battered as flows land. Appadurai and Castells concur that the nodes where these complex processes are most vividly at play are so-called global cities, as we shall see.

Another characteristic of the space of flows relates to time: time is morphed by the space of flows, creating what Castells calls timeless time (see later). Commentators such as David Harvey (1989) – who Castells does cite – have referred to this as the annihilation of space by time, or time–space compression. Instantaneous global communications flows make space shrink, arguably to nothing. Our experience of space, of distance, is profoundly mediated by time: journey time makes us feel how near or far places are, dependent on mode of transport. The world feels smaller from a jet plane than it does on foot. The same goes for communication: a letter to the USA takes days to arrive, an email the blink of an eye (or a cursor). So the space of flows has a set of temporalities, or rhythms, too: fast flows leave jetstreams,

slow flows leave snail trails. And nodes can be ‘sticky’ too, attempting to slow down the flows that land (of course, alternatively they can be shiny, so flows bounce off).

But the space of flows isn’t just this chaotic globe of frantic flows landing, bouncing and swirling overhead; while Castells says at one point that the space of flows ‘connects places as nodes of networks of instrumentality’ and that ‘these places are not meaningful in themselves, but only as nodes of these networks’ (Castells and Ince 2003: 57), where they land, how they land, and how they leave again does matter, is meaningful. But places are only places in so far as they are nodes; like workers, places can be switched off, routed round. The connection is very clear when we consider labour, in fact, given Castells’ aphorism that capital is global but labour is local: the ‘nodeness’ of places is in part a function of the labour available there, both self-programming and generic. Flows will come if the conditions are right, but they can just as easily leave if they are wrong. As Castells (2000b: 10) succinctly puts it, ‘Globalization is highly selective.’ Moreover, this new geography is about the power of networks, not networks of power.

NETWORK CITIES

As noted, so-called global cities are prime nodes or hubs in the space of flows. This raises for Castells, and for us, an interesting conundrum: given the disembedding qualities of the network society, given that time-space compression ‘frees’ people from the constraints of geography, and given all the futurology about new ways of ‘wired living’, why do cities still exist? Even more than exist, in fact: the world is becoming more urbanized (see Graham 2003). Part of the answer comes back to the network logic: the network needs nodes, otherwise it would be pure flow. Castells draws on his previous work with Peter Hall on ‘technopoles’ (Castells and Hall 1994) as part of his explanation, noting that ‘milieux of innovation’ tend to be clustered around urban centres, for a number of reasons, including those management buzz words, synergy and added value. This phenomenon has attracted widespread academic and policy-making attention recently, as people have tried to unlock the magic formula that makes some cities more attractive to innovators and creatives – who are seen as the saviours of cities ravaged by the shift to post industrialism (or informationalism) that gutted urban centres across the world, and also as the engines of a new inter-urban competitiveness, as cities promote themselves to attract wealth and talent (Florida 2002).

There’s also a chicken-and-egg dimension to the equation, in that innovative and creative milieux attract other innovators and creatives, other self-programmable workers. And the infrastructure that builds up to service these milieux – the consumption spaces, taste cultures and so on – are attractive, too (Jayne 2005). As Castells puts it, the global city ‘tends to generate a style of architecture, a certain type

of cosmopolitan aesthetics, and a series of facilities that characterize the lifestyles of the global elite' (Castells and Ince 2003: 57) – restaurants serving *sashimi*, for example, or the assorted 'VIP spaces' that offer a classy welcome. So the space of flows is a network of places that are 'connected around one common, simultaneous social practice' (ibid.: 56): not just the node of electronic communication, but also the site of a constellation of economic, social, cultural and political activities.

Castells makes a number of other insightful comments on the global city in the network society. One is that 'The global city is made of many bits and pieces of cities around the world' (Castells and Ince 2003: 57); another is that 'The global city is not a place, but a process' (Castells 1996a / 2000: 286). Part of this process is about disembedding: global cities emphasize their interconnection with other global cities, and downplay their own proximate hinterlands, leading to what Castells calls an intrametropolitan dualism (see below). Some global cities are mono-functional, but in some senses this no longer matters as they are patched into a network that constitutes the whole. However, like previous mono-industrial cities, the dominance of a single function can render them fragile, subject to the changing flows that may bypass them to find better functionality elsewhere. Inter-urban competitiveness and intra-urban dualism are the twin features of the global network city, therefore. The space of flows impacts on urban form as well as function, Castells argues, simultaneously producing concentration and decentralization: unending urban, suburban and exurban sprawl, edge cities and megacities are variants on what he calls the informational city; these new urban forms, he adds, mean that 'the great urban paradox of the twenty-first century is that we could be living in a predominantly urban world without cities' (Castells 2005: 57) – without what we could hitherto recognize as cities, that is.

Megacities, which he later renames mega-metropolitan regions (Castells 2005), with populations in excess of 10 million, are the primary nodes of the global economy: 'concentrating the directional, productive, and managerial upper functions of all power over the planet; the control of the media; the real politics of power; and the symbolic capacity to create and diffuse messages. ... Megacities articulate the global economy, link up the informational networks, and concentrate the world's power' (Castells 1996a / 2000: 403 – 4). They are globally connected and locally disconnected, often agglomerations of cities and towns of various sizes. An exemplar is China's Pearl River Delta region, soon to be even more mega, Castells argues, as it morphs into the Hong Kong-Shenzhen-Canton-Pearl River Delta-Macau-Zhuhai metropolitan regional system, with a population well over 50 million. Embarking on futurology once more, Castells predicts that such agglomerated mega-metropolitan regions will become the most representative urban form of the twenty-first century, even though they have their own significant and growing problems, such as inadequate transport infrastructure, given that people still commute rather than telecommute (even if their commuting is electronically enhanced, through the use of laptops,

BlackBerrys, mobile phones, etc.). What we are witnessing, Castells (2001b) writes, is ‘the emergence of multi-modal metropolitan mobility’, as nomadic workers shuttle back and forth; the individualization of working patterns, discussed earlier, only serves to heighten this, as work seeps into other parts of our lives. Networked mobility comes to characterize everyday life in the megacity: ‘moving physically while keeping the networking connection to everything we do is a new realm of the human adventure, one about which we know little’ (Castells 2005: 54).

DUAL CITIES

As already mentioned, a stark feature of contemporary urban form – and one predicted to become heightened in the future – is the intrametropolitan dualism. The global city is many cities at once: Sassen (1999) writes that each global city simultaneously contains multiple cities: the corporate, the postindustrial, the third world – layered over each other, and often in conflict. Moreover, Castells highlights another key spatial dualism, between the space of flows and the space of places: ‘The space of flows does not permeate down to the whole realm of human experience in the network society. Indeed, the overwhelming majority of people, in advanced and traditional societies alike, live in places, and so perceive their space as place-based’ (Castells 1996a / 2000: 423). Places are localized, rooted, and people derive meaning from that rootedness. While Castells appears to romanticize place, and (in his earlier work especially) to overplay the antinomy between global and local that flows and places embody, he reiterates the crucial point that ‘people do still live in places’ (ibid.: 428), even if the meaning of those places is transformed by the logic of networks and flows.

In later writing and talks, Castells revises his view slightly, to see the space of flows and the space of places as more coterminous, or folded together – he even talks of ‘*cyborg cities*, or hybrid cities made up by the intertwining of flows and places’ (Castells 2005: 54, emphasis in original). He also sees an error in his own prior articulation of the space of flows only to the techno-elites; because the space of flows is now predominantly about electronic communications, ‘people of all kinds, wishing to do all kinds of things, can occupy this space of flows and use it for their own purposes’ (Castells and Ince 2003: 58). The key example he refers to as ‘grassrooting’ the space of flows (Castells 1999) – the use of the network, the use of flows, by social movements, such as anti-globalization protests (Castells 2005). And in *The Internet Galaxy* (2001b), he further points to more varied uses of the space of flows to constitute a new public space, or agora, for open discussion of ... well, of anything and everything. Some of this is ‘political’, some of it is ‘progressive’. As he had argued before, the fact that some network content is far

from progressive should not be used as a justification for tighter control or censorship on the net: 'The internet brings us face to face with the mirror of who we actually are. So I would rather work on ourselves than close down the net' (Castells 1999: 298).

There is another powerful urban dualism to consider in the network society: 'the simultaneous growth and decline of economies and societies within the same metropolitan area' (Castells 2002 [1996]: 132). Now, this is not simply the dualism of the space of flows and the space of places, as Castells had earlier conceived it. Things are more complex than that, and the relationships between growth and decline are still being mapped and analysed. Socioeconomic polarization between cities and within cities is the key point to remember: 'splintering urbanism' is the hallmark of the network society (Graham and Marvin 2001). Perhaps we should talk of 'glocal cities', then: at once global and local (Castells 2001b).

Indeed, in his most recent work, Castells turns his full focus to the city in the age of information, diagnosing its maladies and suggesting remedies. What is being lost in mega-urban sprawl and flow is 'the culture of cities', the role of cities as sites of cultural communication (Castells 2005). Public spaces 'as sites of spontaneous social interaction' are needed, not to encourage assimilation but to promote 'sharing the city by irreversibly distinct cultures and identities' (ibid.: 52) – a common theme in recent commentaries on the future of cities (e.g. Amin and Thrift 2002). Castells' programme for urban rejuvenation calls for a focus on communication in the broad sense:

Restoring functional communication by metropolitan planning; providing spatial meaning by a new symbolic nodality created by innovative architectural projects; and reinstating the city in its urban form by the practice of urban design focused on the preservation, restoration, and construction of public space as the epitome of urban life.

(Castells 2005: 59)

TIMELESS TIME

Space and time are closely related in the logic of networks and flows. In place of Harvey's (1989) time - space compression, where space is 'annihilated' by time, or Giddens' (1991) time - space distancing, where the link between time and space is decoupled, Castells talks of the space of flows and timeless time. This means a number of things. First, and most obviously, it refers to speeding up. Making things faster is one of the most important imperatives of the information age (Erkisen 2001), so much so that it has birthed its own countermovement or cultural commune, in 'slow living' (Parkins and Craig 2006). So instantaneity is one form of timeless time. Another is called by Castells desequencing: as a result of living in a multimedia age with limitless access to streams of live and archived material, as

well as ever more wondrous ways to predict or imagine the future, we are exposed to a montage of instants wrenched from temporal context: past, present and future are disassembled and reassembled for us and by us (see below). Without the anchoring of temporality, we live, as some postmodern commentators argue, in a 'perpetual present': the future arrives almost before we've thought of it, the past comes back at us in soundbites: 'we live ... in the encyclopedia of historical experience, all our tenses at the same time, being able to reorder them in a composite created by our own fantasy or our interests' (Castells 2002 [1996]: 131). Greater capacity for archiving our own lives, for example through digital photography, only enhances this experience, as we review and edit the masses of pasts we collect and store (Van Dijk 2005).

Outside of personal experience, timeless time is evident for Castells in split-second financial transactions on the global market, in so-called 'instant wars' comprising surgical strikes, and in the technological transformation of the lifecycle, through new reproductive technologies, antiageing and even, through cryogenics, the idea of deferring death. So we have a curious mix: the culture of the ephemeral and of the eternal. Castells calls this the breaking down of 'rhythmicity', and he is strikingly critical of its manifestations and motivations. He is, of course, also mindful that most people do still live by biological time and clock time; as with his discussion of the space of flows, timeless time primarily characterizes dominant social groups and functions in the network society. Yet variants are experienced outside of elites, for example in television's endless recycling of its own past, in 'retrofuturism', in the growing popularity and availability of aesthetic surgery procedures, or in the policy concept of 'work-life balance'.

SUMMARY

The space of flows is Castells' conceptualization of the geography of the network society: a global array of disjunctive flows, of different things (but increasingly of electronic things and knowledge things) that move around the world, emanating from and landing in nodes, i.e. places. Global cities are the prime movers of the space of flows, and likely to be more so as they morph into megacities or metropolitan regions, ironically the ideal urban form of the information age. But cities are riven by contradiction and duality: social polarization is getting worse. Where he used to see an opposition between the dominant space of flows and the switched-off space of places, Castells now presents a more complex picture, due in part to the 'grassrooting' of the space of

flows by social movements. Time as well as space is warped by the network society, at least for dominant groups and functions. It is accelerated, randomized and desquenced, leading to timeless time and temporal perturbations that upset the former rhythms of life.

3 REAL VIRTUALITY

Real virtuality, Castells' easy pun on virtual reality, opens up a space to discuss the culture of the network society. As might be expected, this is a culture heavily shaped by the media; only the old-fashioned mass media, feeding its mass audience a standardized product,